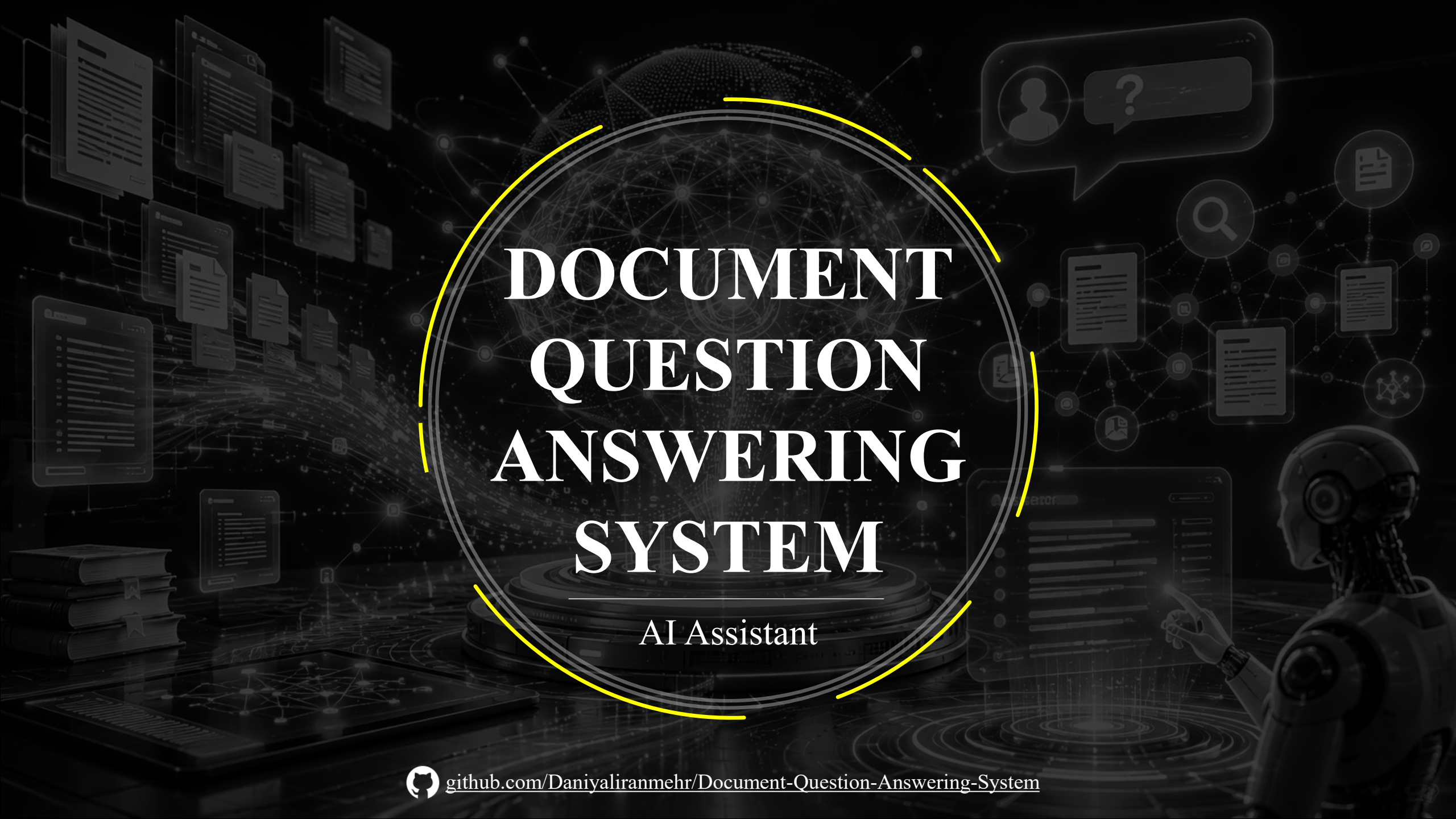




DOCUMENT QUESTION ANSWERING SYSTEM

AI Assistant



github.com/Daniyaliranmehr/Document-Question-Answering-System

1



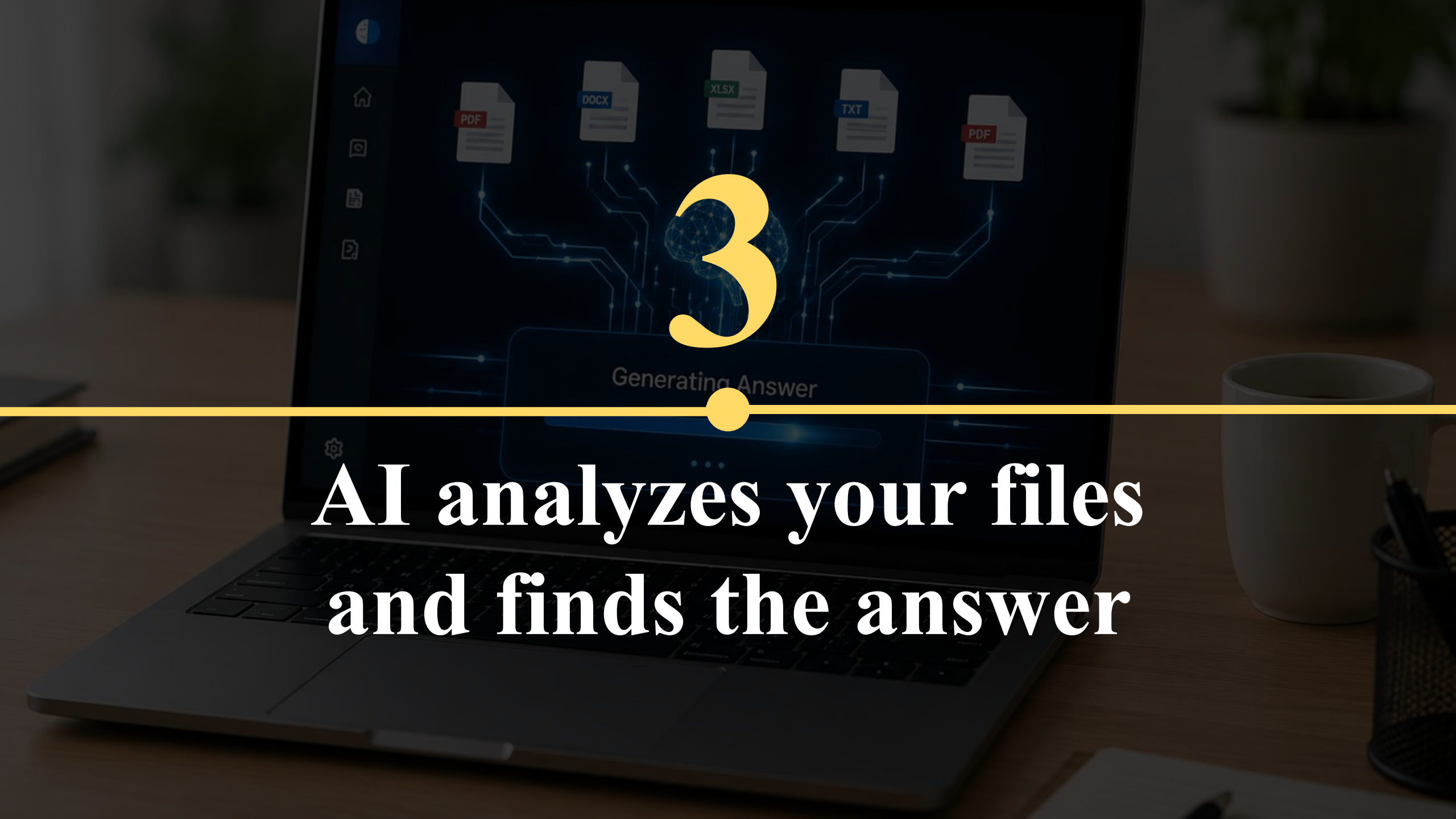
Upload your files



A person is seen from the side, sitting at a desk and using a laptop. The laptop screen displays a chat interface with a header 'Ask a Question' and a text input box containing the question 'What are the main findings of this research?'. A large yellow number '2' is overlaid on the screen. A horizontal yellow line with a dot in the center runs across the middle of the image.

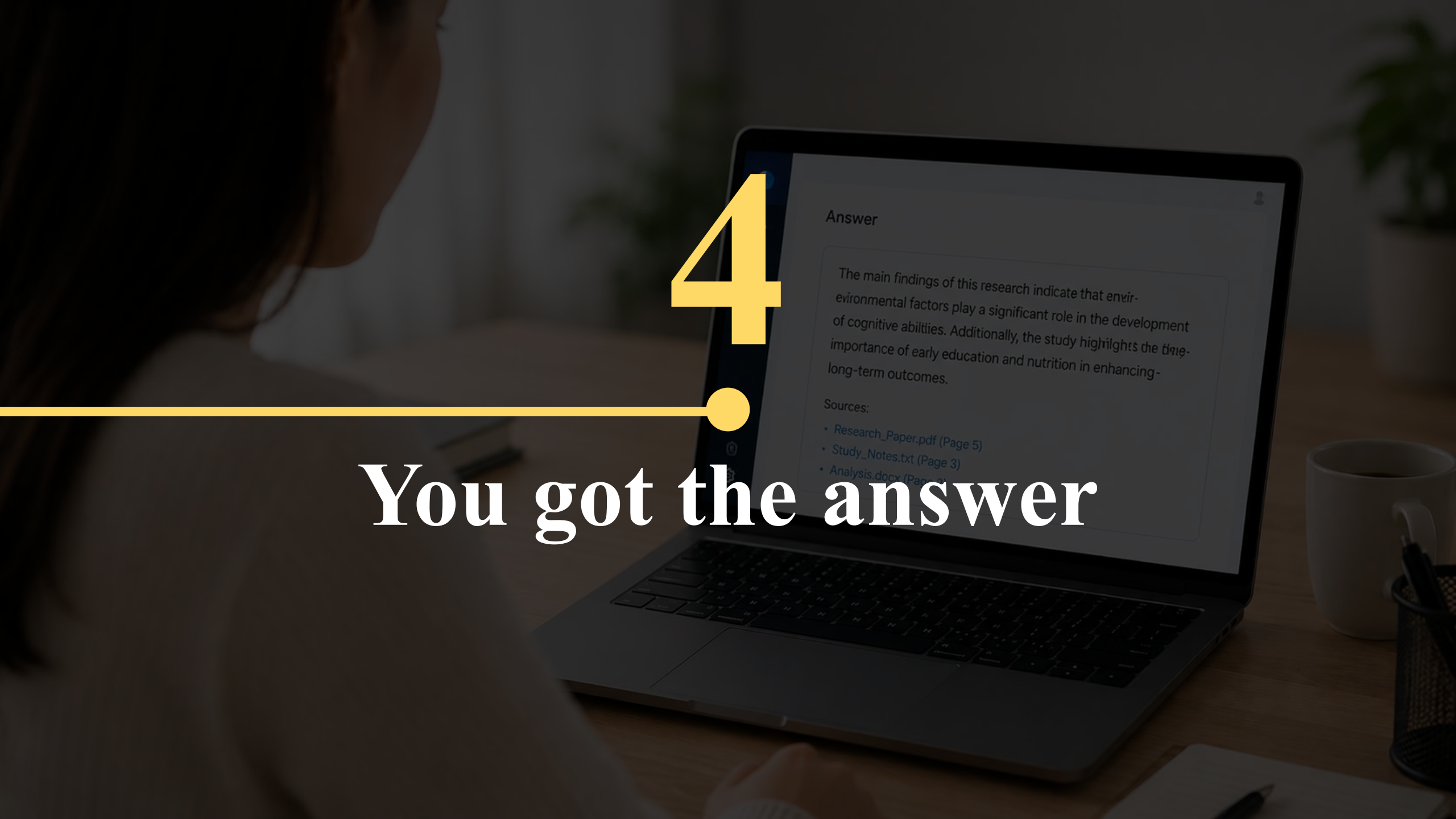
2

Ask a question



3

**AI analyzes your files
and finds the answer**

A person is seen from the side, looking at a laptop screen. A large, bold yellow number '4' is overlaid on the image, with a horizontal yellow line extending from its base across the middle of the frame. The laptop screen displays a document titled 'Answer' with a paragraph of text and a list of sources. The background is slightly blurred, showing a desk with a cup and some papers.

4

You got the answer

Applications

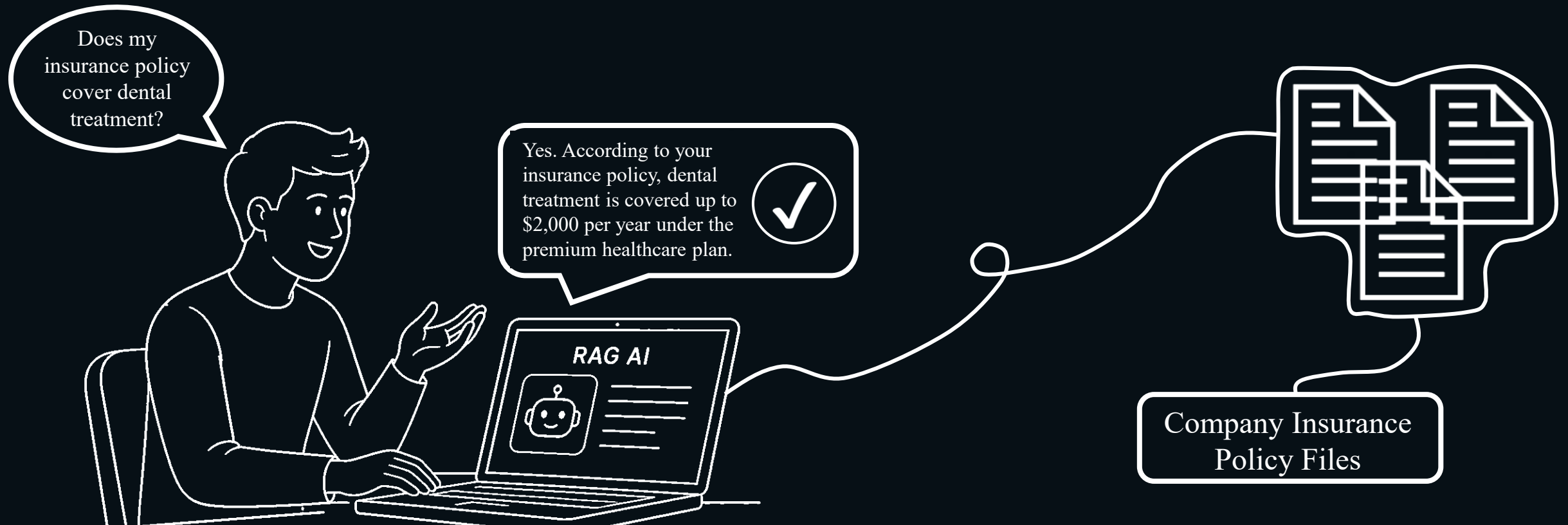
At its core, this system is industry-agnostic. In today's digital era, organizations across every sector are drowning in data but starving for actionable knowledge. Whether it's legal compliance, healthcare research, corporate knowledge management, or financial analysis, the core challenge remains the same: extracting precise answers from massive volumes of text instantly.

Instead of reinventing the wheel for each sector, this RAG-powered architecture serves as a universal engine. Let's explore how this technology seamlessly adapts to different domains and solves real-world bottlenecks.



Customer Support

As a Customer Support Manager, you can upload your entire product documentation into the system to provide lightning-fast, 24/7 automated responses, drastically reducing the pressure on your support team while improving customer satisfaction.



Customer Support



24/7 Instant Resolution

Eliminate customer wait times by providing immediate, accurate answers around the clock.



Massive Ticket Deflection

Drastically reduce support workload by automatically handling repetitive queries using your existing documentation.



Hallucination-Free Accuracy

Deliver highly reliable answers grounded strictly in your official product manuals, ensuring users get factual information.

Customer Support

Process	Traditional Method	AI-Powered RAG System
Finding Answers	Agents manually search through scattered PDFs, wikis, and lengthy manuals to find solutions.	The AI instantly scans thousands of documents and generates a precise, context-aware answer.
Handling Volume	High ticket volumes lead to long customer wait times and require hiring more support staff.	Scales effortlessly to handle thousands of concurrent queries with zero wait time.
Knowledge Updates	Support teams must be manually retrained whenever a new feature or manual is released.	Simply upload the new document; the AI immediately learns and applies the updated information.
Customer Experience	Users are forced to navigate complex FAQ pages or wait hours for an email reply.	Users ask natural language questions and receive conversational, direct answers instantly.

Legal Document Analysis

As a Lawyer, you can query your firm's entire private database of confidential cases and active contracts with complete data privacy, receiving answers that are strictly grounded in your uploaded files. Unlike generic AI tools, this system completely eliminates the risk of hallucination and provides direct citations to the exact page and paragraph of the document, allowing you to instantly verify every single fact before presenting it in court.



Legal Document Analysis



Accelerated Due Diligence

Analyze complex, multi-page contracts in seconds to extract critical clauses and obligations instantly.



Bulletproof Risk Mitigation

Minimize human oversight by scanning entire legal documents for hidden liabilities, strict deadlines, or non-compliance issues.



Maximize Efficiency

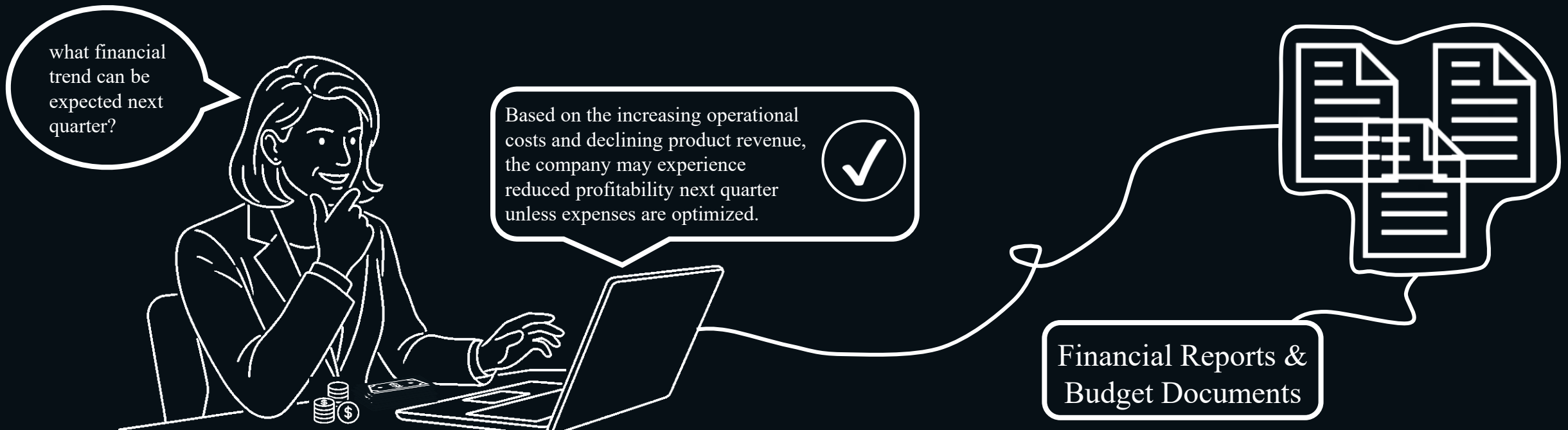
Shift legal professionals from tedious, hours-long document skimming to high-value strategic consulting and case preparation.

Legal Document Analysis

Process	Traditional Method	AI-Powered RAG System
Clause Extraction	Lawyers must manually flip through hundreds of pages of legal jargon to find specific termination or payment terms.	The AI instantly locates, extracts, and summarizes exact terms via simple natural language queries.
Review & Turnaround Speed	Reviewing a new contract or NDA takes hours or days, creating operational bottlenecks and delaying business deals.	Contracts are analyzed and vetted instantly, dramatically accelerating deal cycles and contract approvals.
Cross-Document Analysis	Comparing historical contracts or cross-referencing multiple cases requires extensive manual cross-checking.	Simply upload the new document; the AI immediately learns and applies the updated information.
Information Extraction	Requires senior legal expertise just to locate and interpret basic factual data scattered across archives.	Empowers paralegals or junior staff to retrieve verified factual data instantly, reserving senior hours for strategic work.

Financial Auditing & Compliance

As a Chief Financial Officer, you can instantly query your organization's entire repository of private tax codes, audit histories, and internal financial statements to extract precise compliance answers with absolute data security. Unlike public AI tools that risk leaking sensitive corporate numbers, this system locks down your financial data and delivers verified insights backed by direct document citations, turning complex financial auditing into a fast and secure process.



Financial Auditing & Compliance



Lightning-Fast Audit Preparation

Locate specific transactions, ledger anomalies, or historical financial data across thousands of pages in seconds.



Automated Regulatory Compliance

Instantly scan evolving tax laws and financial regulations to guarantee absolute compliance and avoid costly penalties.



Leak-Proof Financial Security

Process highly confidential balance sheets and corporate invoices safely within a secure environment with zero external data exposure.

Financial Auditing & Compliance

Process	Traditional Method	AI-Powered RAG System
Tax Law Research	Finance teams spend days manually cross-referencing thousands of pages of dynamic, confusing tax codes.	The AI analyzes the dynamic regulations instantly and gives exact answers with direct code citations.
Discrepancy Spotting	Auditors must manually flip through endless receipts, invoices, and ledger books to verify transactions.	Simply query the system to pull up any transaction, missing link, or financial record within seconds.
Data Privacy & AI	Financial data cannot be uploaded to other tools due to strict privacy laws and risk of leaks.	Operates in a completely isolated architecture, protecting proprietary numbers while utilizing advanced AI.

Scientific Research

As a Researcher, you can leverage a system where the database is pre-populated with a curated collection of scientific papers specifically within your field of study. Instead of dealing with generic language models that pull raw information from the vast internet, you query a targeted knowledge base of domain-specific literature. This ensures that every extracted methodology, dataset, or finding is hyper-focused and strictly derived from the exact body of research relevant to your work.

How do recent multi-view clustering methods handle incomplete views and missing data?

Recent multi-view clustering approaches address incomplete views by learning shared latent representations and using cross-view consistency constraints. Several papers apply ...



Scientific Papers on
Multi-View Clustering



Scientific Research



Cross-Paper Domain Synthesis

Query dozens of specialized papers in a specific field simultaneously to instantly synthesize trends and extract core methodologies across the entire domain.



Target-Locked Information Mining

Eliminate irrelevant web-search noise by confining the AI's knowledge strictly to your uploaded corpus, capturing deep technical data that generic AIs miss.



Research Gap & Novelty Discovery

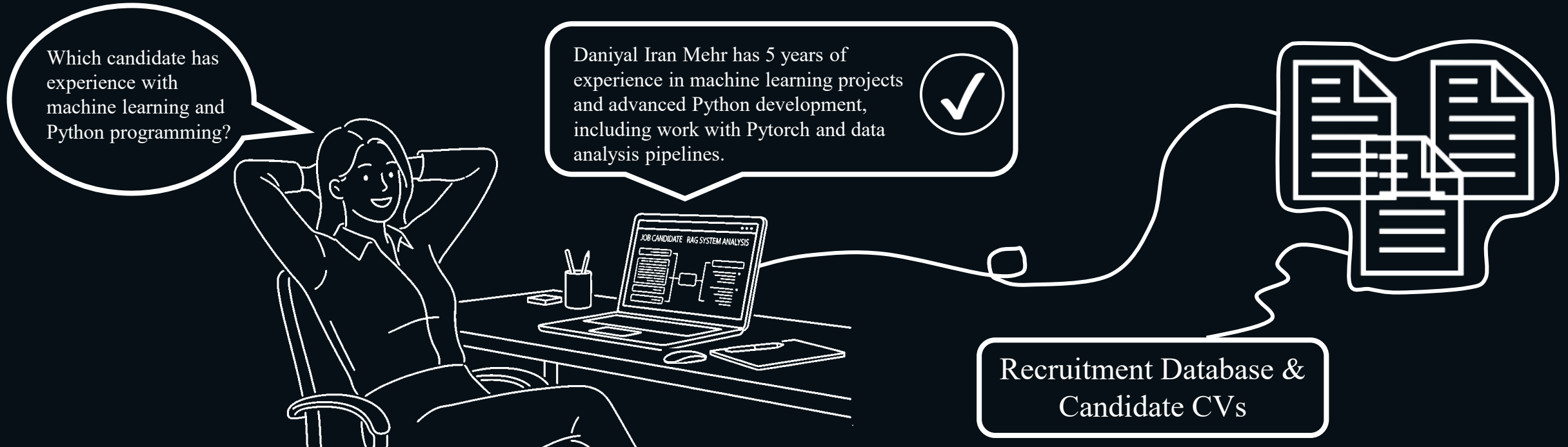
Identify research gaps and unresolved challenges across papers to discover new opportunities for innovation and publication.

Scientific Research

Process	Traditional Method	AI-Powered RAG System
Knowledge Source	Relies on scattered academic databases and broad web searches, resulting in fragmented, imprecise, or contextually disconnected information	Operates strictly on a pre-loaded database of curated, peer-reviewed papers for your specific field..
Data Boundary	No boundaries; answers can come from any unchecked online source, leading to high hallucination risks.	Strictly bounded within the targeted repository, eliminating the risk of outside text contaminating the results..
Literature Synthesis & Gap Isolation	Requires reading papers sequentially and mapping connections manually via spreadsheets, a slow process prone to overlooking hidden contradictions or missing links between different studies.	Uses parallel processing to cross-reference the pre-loaded repository instantly, highlighting exactly where different authors disagree, align, or leave an unexplored research gap.

HR & Recruitment Systems

As an HR Director, you can radically accelerate hiring and internal operations by turning static resumes and dense policies into a dynamic knowledge base. Instead of spending weeks manually screening applicants or digging through lengthy handbooks, simply ask natural language questions to instantly surface the perfect candidate or precise guidelines. This system securely isolates your proprietary data, ensuring 100% privacy for rapid, data-driven workforce decisions.



HR & Recruitment Systems



Faster Recruitment Decisions

Reduce recruitment time by automatically extracting and comparing candidate skills, experiences, and qualifications from uploaded CVs.



Semantic Talent Matching

Instantly scan hundreds of uploaded resumes to extract actual context, project depth, and hidden capabilities of candidates beyond mere keyword matching.



Centralized HR Knowledge Access

Allow employees and HR teams to instantly access company policies, onboarding procedures, and internal guidelines through conversational AI.

HR & Recruitment Systems

Process	Traditional Method	AI-Powered RAG System
Applicant Screening	Relies on manual reading or rigid keyword-based ATS software, frequently rejecting highly qualified candidates who used different synonyms for a skill.	Uses deep semantic understanding to evaluate a candidate's actual experience context, accurately matching skills regardless of the specific terminology used.
Candidate Comparison	Recruiters manually extract applicant data into spreadsheets to compare backgrounds, a slow process heavily prone to human bias, fatigue, and oversight.	Simultaneously cross-references multiple uploaded resumes against the job description, generating a structured, unbiased skills and experience comparison instantly.
HR Policy Navigation	Employees and managers waste hours searching through outdated, multi-page PDF handbooks or submitting internal support tickets for simple protocol questions.	Acts as an instant internal HR assistant, extracting precise clauses and step-by-step procedures directly from verified corporate guidelines in seconds.

Core Technology Infrastructure



LangChain



Django



RAG

Detailed technical documentation is available in the repository's README:



github.com/Daniyaliranmehr/Document-Question-Answering-System

The Project is not Closed!

It is still under development...





Daniyal Iran Mehr

Artificial Intelligence | Data Science

Hi there! I'm a computer engineer passionate about Artificial Intelligence and Data Science. I truly believe this field is shaping our future, and I'm on a journey to continuously learn and make a meaningful impact. Feel free to follow my work if you're interested. I'm always excited to connect and collaborate with like-minded people in this space.



github.com/Daniyaliranmehr



linkedin.com/in/daniyaliranmehr



daniyaliranmehr@gmail.com